Adam Danicki he/him/his

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SUMMARY

Research data analyst and Computer Science graduate with experience building Python-based analysis workflows, reproducible modeling pipelines, PostgreSQL-backed data systems, validation checks, and reporting-ready outputs. Strong foundation in statistics, machine learning, data quality review, technical documentation, Python, SQL, pandas, NumPy, Matplotlib, scikit-learn, PostgreSQL, FastAPI, SQLAlchemy, and Docker. Current healthcare experience using Epic EHR workflows to monitor, document, and validate clinical data in a hematology/oncology setting, with interest in applying computational methods to healthcare research, disease modeling, and population health analysis.

EDUCATION

University of Massachusetts Amherst | College of Information and Computer Sciences

Bachelor of Science in Computer Science, May 2025 | GPA 3.408 (Spring 2025 term GPA 3.936)

Relevant coursework: Statistics, Machine Learning, Algorithms, Software Engineering, Operating Systems, Computer Networks, Linear Algebra, Computer Systems Principles, Programming with Data Structures, Reasoning Under Uncertainty

PROJECTS

CICS Degree Planner | React, TypeScript, HTML/CSS, API, Docker

- Developed React and TypeScript components for a multi-semester degree planning platform with course selection, schedule visualization, and interactive planning workflows.
- Integrated structured course data into frontend state and reusable UI components while maintaining responsive behavior during frequent user interactions and state changes.
- Collaborated on feature design, debugging, code integration, and iterative delivery within a 6-person Agile team using Docker-based development environments.

Custom ML Benchmarking Suite | Python, NumPy, pandas, Matplotlib, scikit-learn

- Built a reusable benchmarking framework evaluating k-NN, Random Forests, and Neural Networks across multiple datasets using normalization, k-fold cross-validation, and consistent metric reporting.
- Implemented preprocessing and evaluation workflows for mixed numerical and categorical features, producing accuracy/F1 summaries and Matplotlib figures for technical analysis.
- Achieved strong model results including 98.55% accuracy with k-NN, 93.05% with Random Forests, and 92.8% with Neural Networks.

Longitudinal Lesion Tracking REST API | Python, FastAPI, SQLAlchemy, PostgreSQL, Docker Compose

- Built a Dockerized PostgreSQL-backed REST API with 26 endpoints supporting CRUD operations, nested read views, validation logic, and analytics-style queries for structured longitudinal measurement records.
- Designed a normalized relational schema with foreign keys, cascading deletes, scoped uniqueness constraints, and validation checks to prevent duplicate or inconsistent records.
- Created FastAPI/OpenAPI documentation and automated demo data workflows to support repeatable testing, endpoint validation, and technical handoffs.

Gesture-Controlled Drone System | Python, SVM, ESP32, IMU, Tello SDK

- Built a real-time sensor-driven control system that streamed IMU data from an ESP32, transformed accelerometer and gyroscope signals into FFT-based features, and triggered Tello drone maneuvers.
- Collected and labeled 420 IMU sequences across 7 gestures and 3 users, trained an RBF SVM achieving 94% multi-class accuracy.
- Implemented and documented a low-latency inference pipeline connecting sensor data to gesture classification and drone actions including movement, rotation, and flips.

SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, HTML/CSS
Systems and analysis: requirements translation, technical documentation, debugging, system testing, performance evaluation, tradeoff analysis
Data and modeling: NumPy, pandas, Matplotlib, scikit-learn, feature engineering, cross-validation, model evaluation, reporting-ready outputs
Hardware/software: ESP32, IMU sensors, sensor-data streaming, real-time inference, device control logic
Backend and APIs: FastAPI, REST APIs, OpenAPI/Swagger, SQLAlchemy, PostgreSQL
Tools: Git/GitHub, Linux, Docker, Docker Compose, Agile/Scrum fundamentals

EXPERIENCE

Patient Care Technician - Hematology/Oncology (Boston, MA) | Beth Israel Deaconess Medical Center — April 2026 - Present

- Monitor, document, and validate patient clinical data in Epic EHR workflows to support patient status tracking, care continuity, and escalation.
- Review patient vitals and condition trends in a high-acuity hematology/oncology environment, identifying abnormal changes and communicating findings to nursing staff.

IT Support Administrator (Boston, MA) | Pope John Paul II Polish Language School — September 2017 - August 2023

- Provided onsite technical support for instructors and staff by troubleshooting classroom computers, Wi-Fi connectivity, display systems, printers, AV equipment, and setup issues during live weekend school operations.
- Supported student onboarding, digital records, and staff technology access workflows for a Saturday language school serving 130+ students.